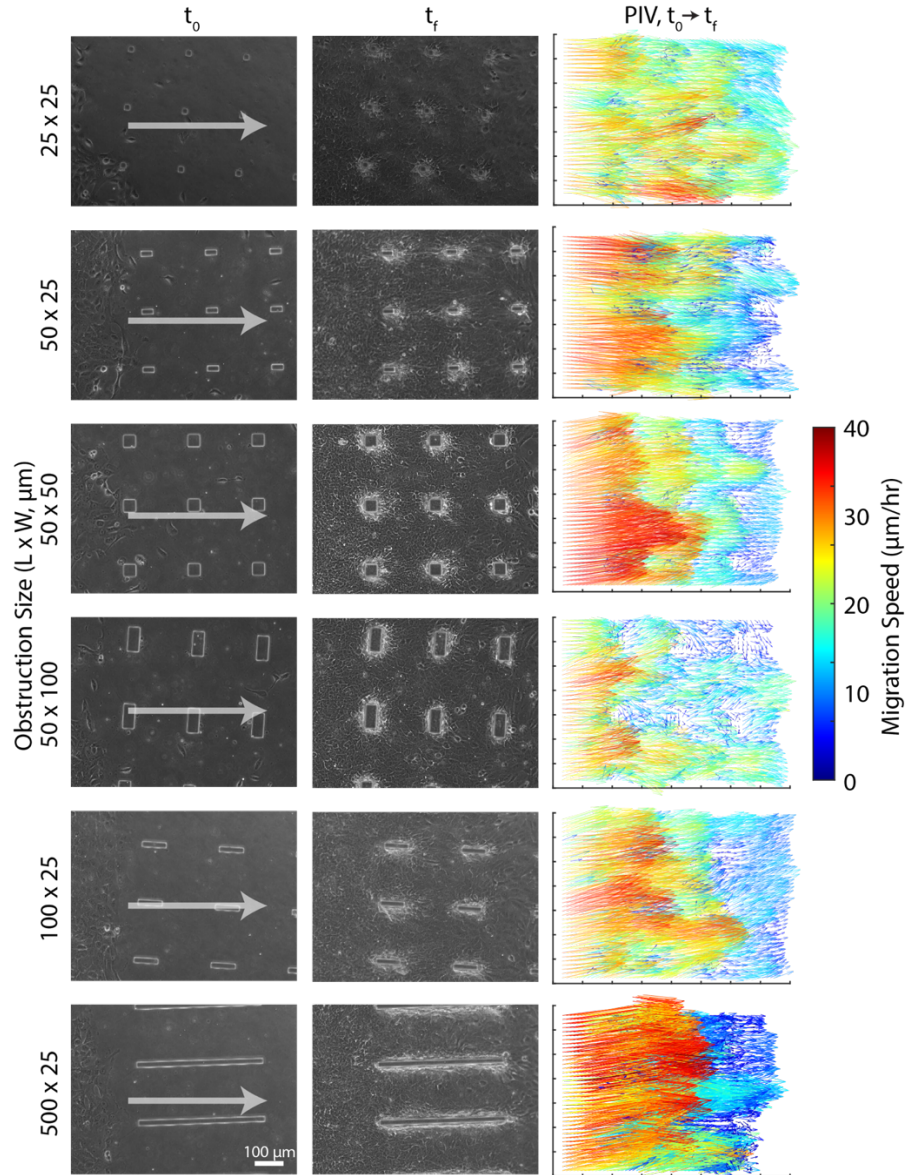
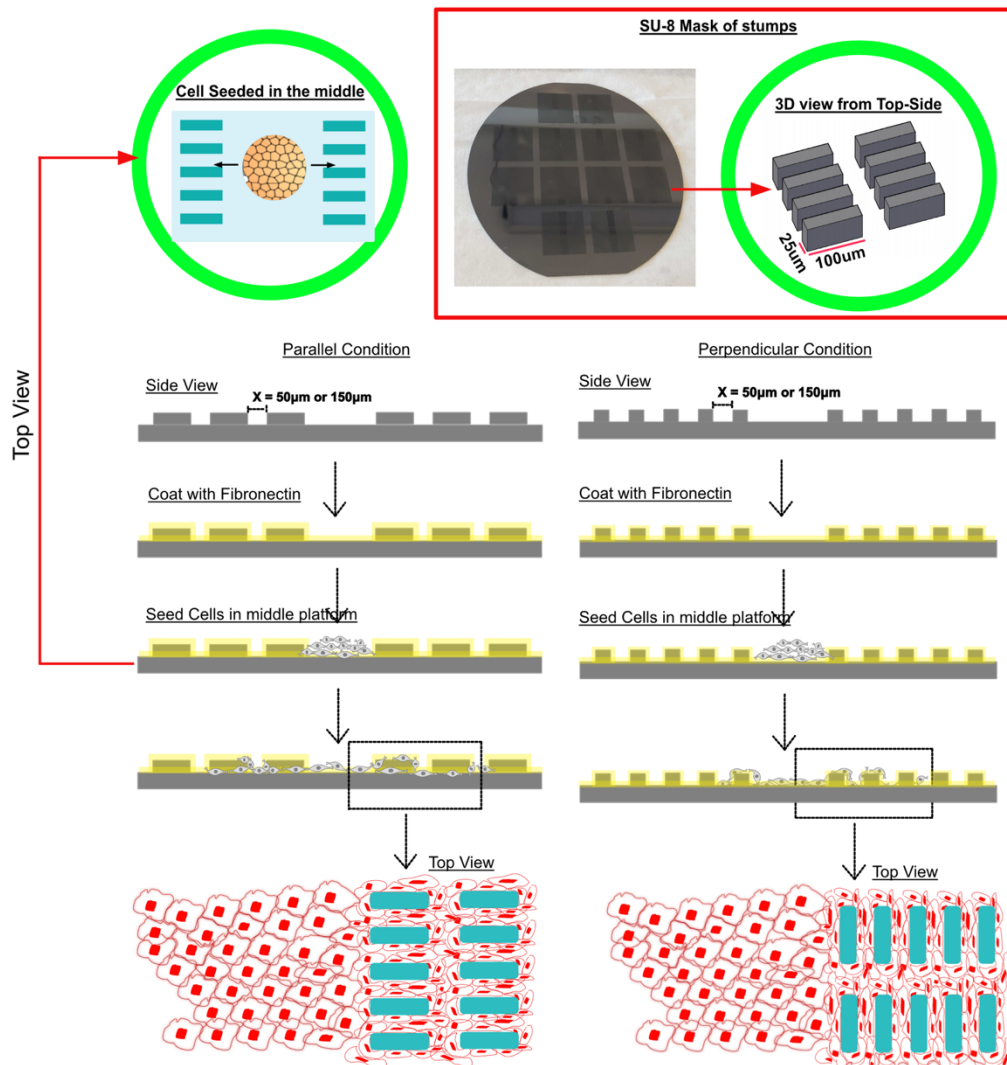


Supplementary Materials
Molecular Biology of the Cell
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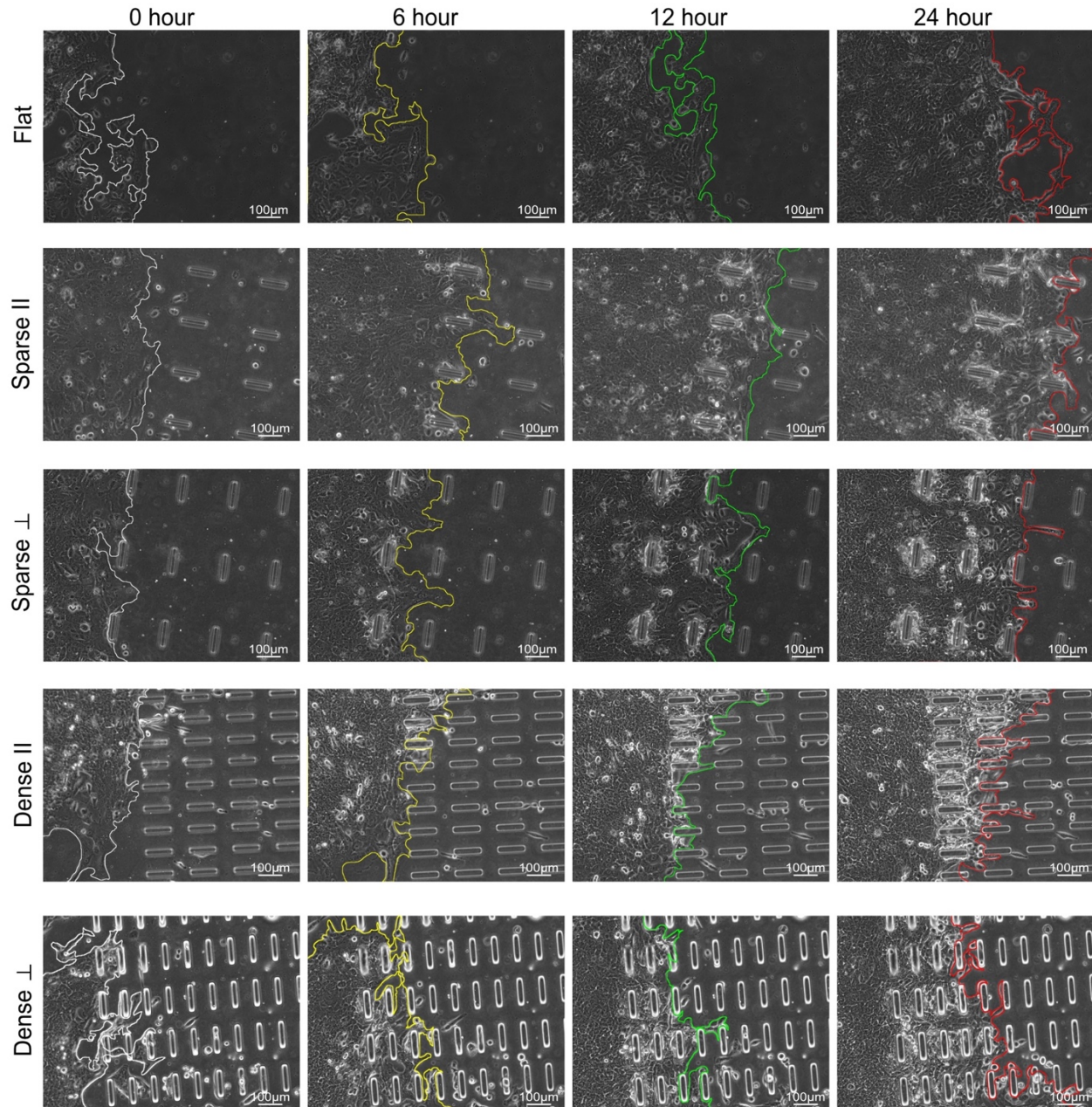
Supplementary Figures



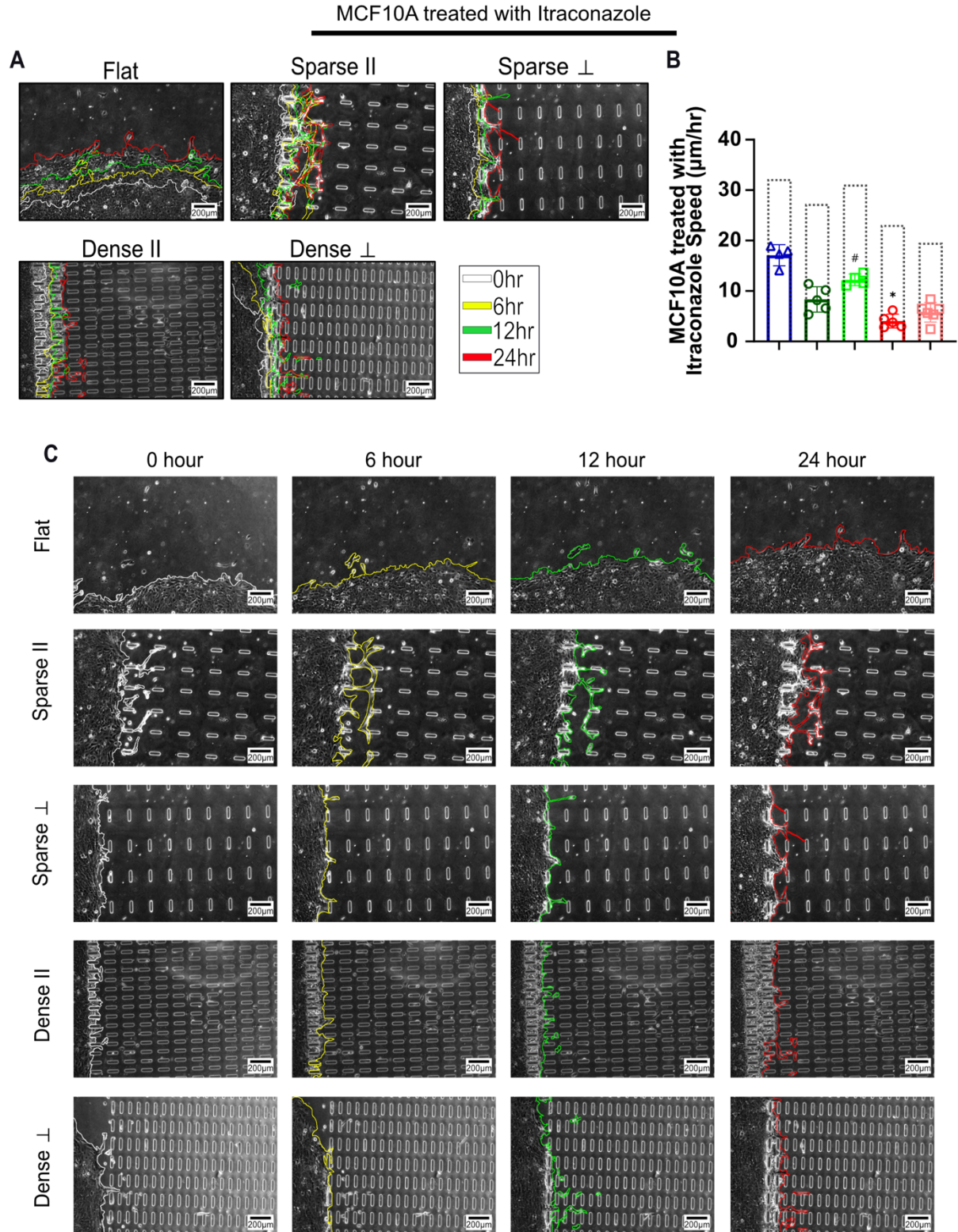
Supplementary Figure 1. Various obstruction configurations to clarify the effect of ECM obstructions on cell migration. (A) The figure shows early obstruction configurations that were utilized to study the effects of ECM obstructions on collective cell migration. Obstruction sizes are labeled such that the length describes the dimension along the direction of migration (horizontal) and width describes the dimension perpendicular to migration (vertical). Images show the initial (t_0) and final (t_f) time points used to track cell migration through the posts. Large arrows on t_0 images indicate direction of migration. Scale bar = 100 μm . PIV vectors (right) indicate direction and magnitude of cell migration speed from t_0 to t_f . It can be seen that varying obstruction size both parallel and perpendicular to migration produces variable responses, adding to the complexity of selecting obstruction shape and size. However, more migration hindrance can be seen when the perpendicular dimension of the obstruction is greater than its parallel dimension (100 x 50 condition).



Supplementary Figure 2. Lithography steps for PDMS substrate preparation. Using hard lithography, a SU-8 mold is fabricated on silicon discs. These masters are used to perform soft lithography and microfabricate PDMS substrates with defined geometry for stumps. Cells are seeded in the middle and outward collective cell migration is analyzed.

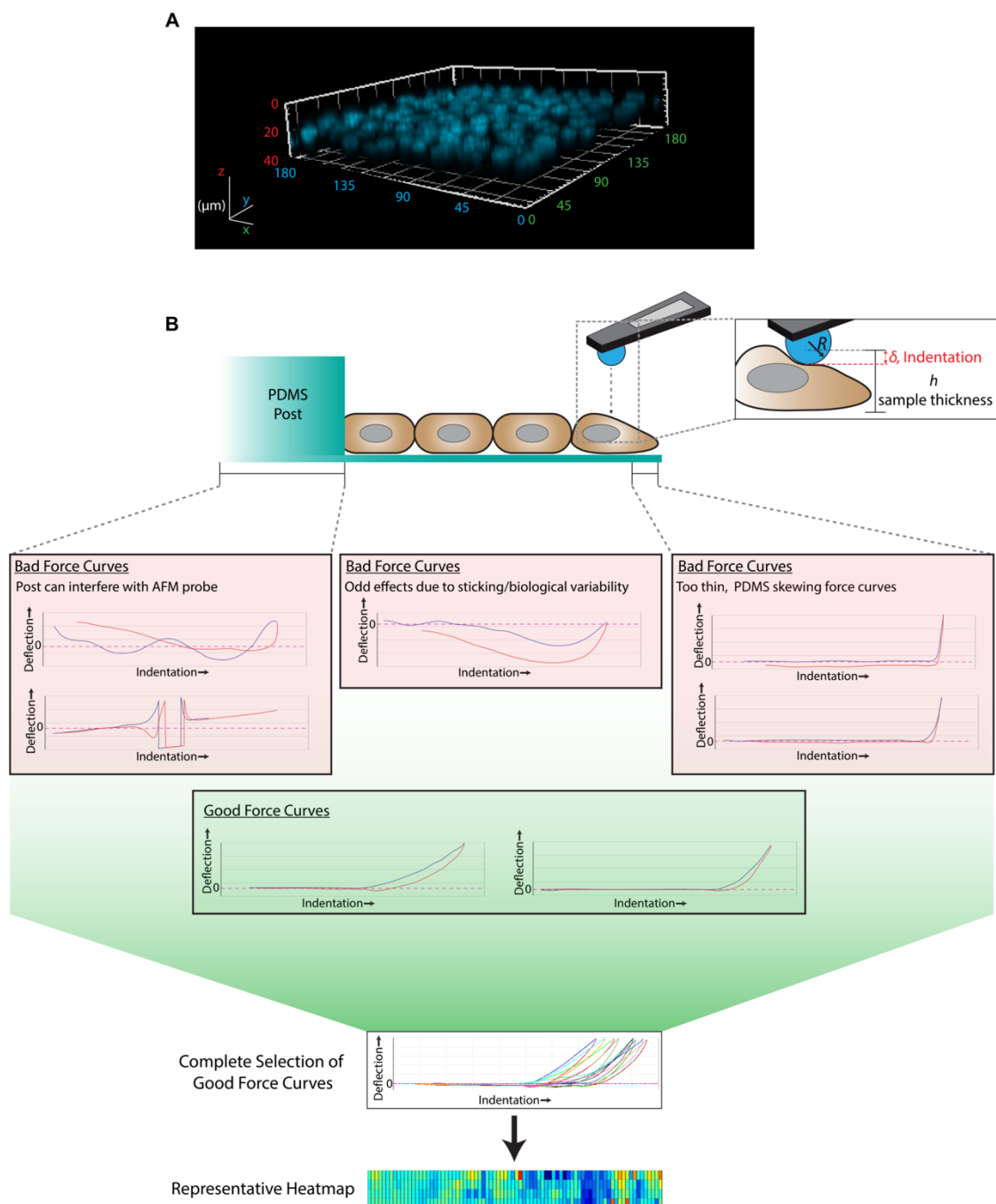


Supplementary Figure 3. Leading edge shapes for MCF10A epithelial monolayers. Representative phase contrast images at different timepoints between 0-24hr at 6hr interval, with annotated leading edge shapes, for 5 substrate conditions of varying stump spacing and orientation. Scale bar = 100μm.

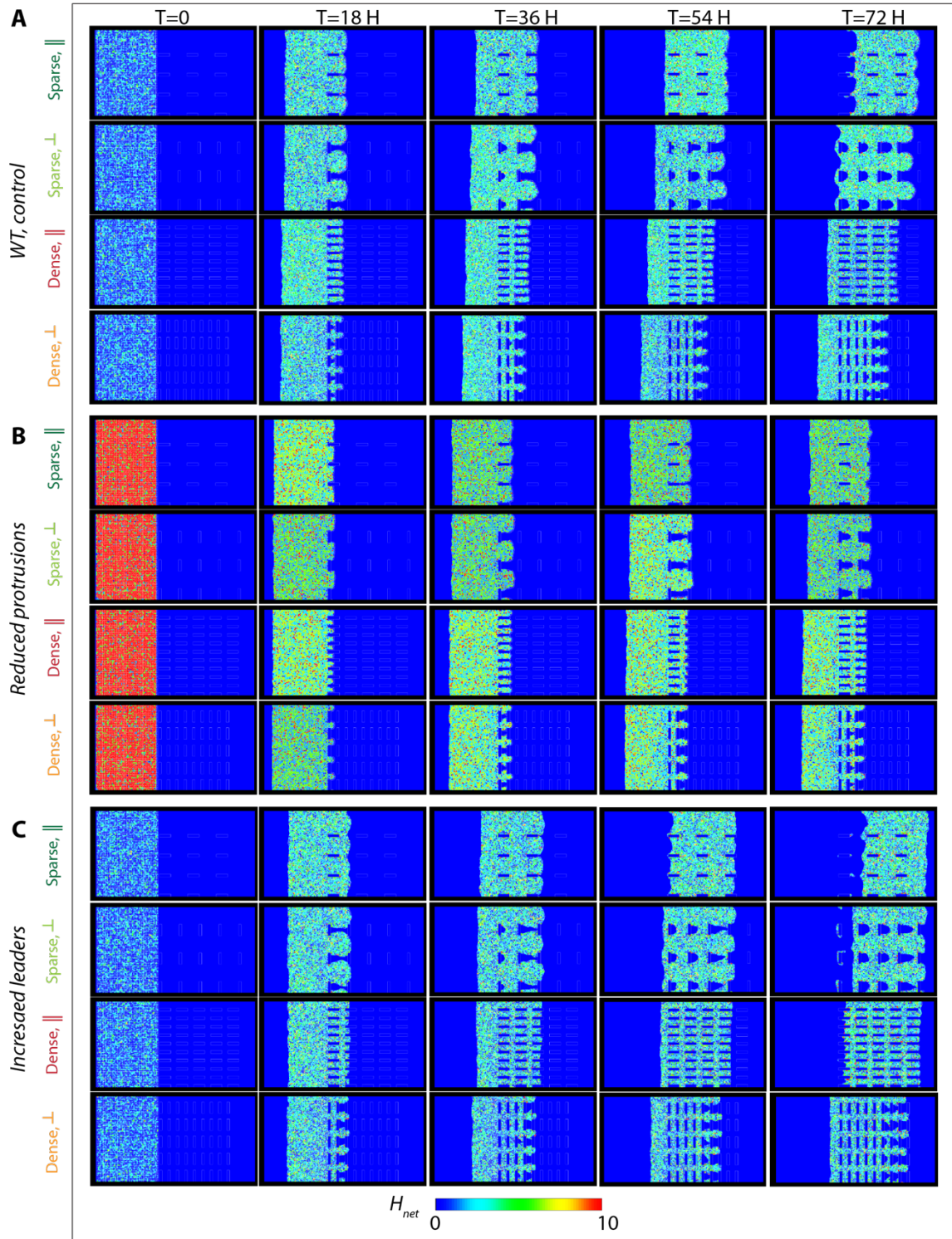


Supplementary Figure 4. MCF10A treated with Itraconazole epithelial monolayers collective cell migration and leading edge shapes. . (A) Representative phase contrast images at 24hr timepoint of MCF10A epithelial cells treated with Itraconazole in four substrate conditions with varying obstruction spacing and orientation. Leading edges of migrating colonies are annotated for timepoints at 6hr interval

between 0-24hr. Migration trajectories of individual cells within Itraconazole treated epithelial monolayers are plotted with color coding for time-averaged **(B)** speed. **(C)** Representative phase contrast images of collectively migrating MCF10A treated with Itraconazole cells at different timepoints between 0-24hr at 6hr interval, with annotated leading edge shapes, for 5 substrate conditions of varying stump spacing and orientation. Scale bar = 200 μ m. Significant difference annotated as * $p < 0.05$ for comparison against the sparse condition, and # $p < 0.05$ for comparison against the parallel condition. $N \geq 5$ fields of view in all plots.

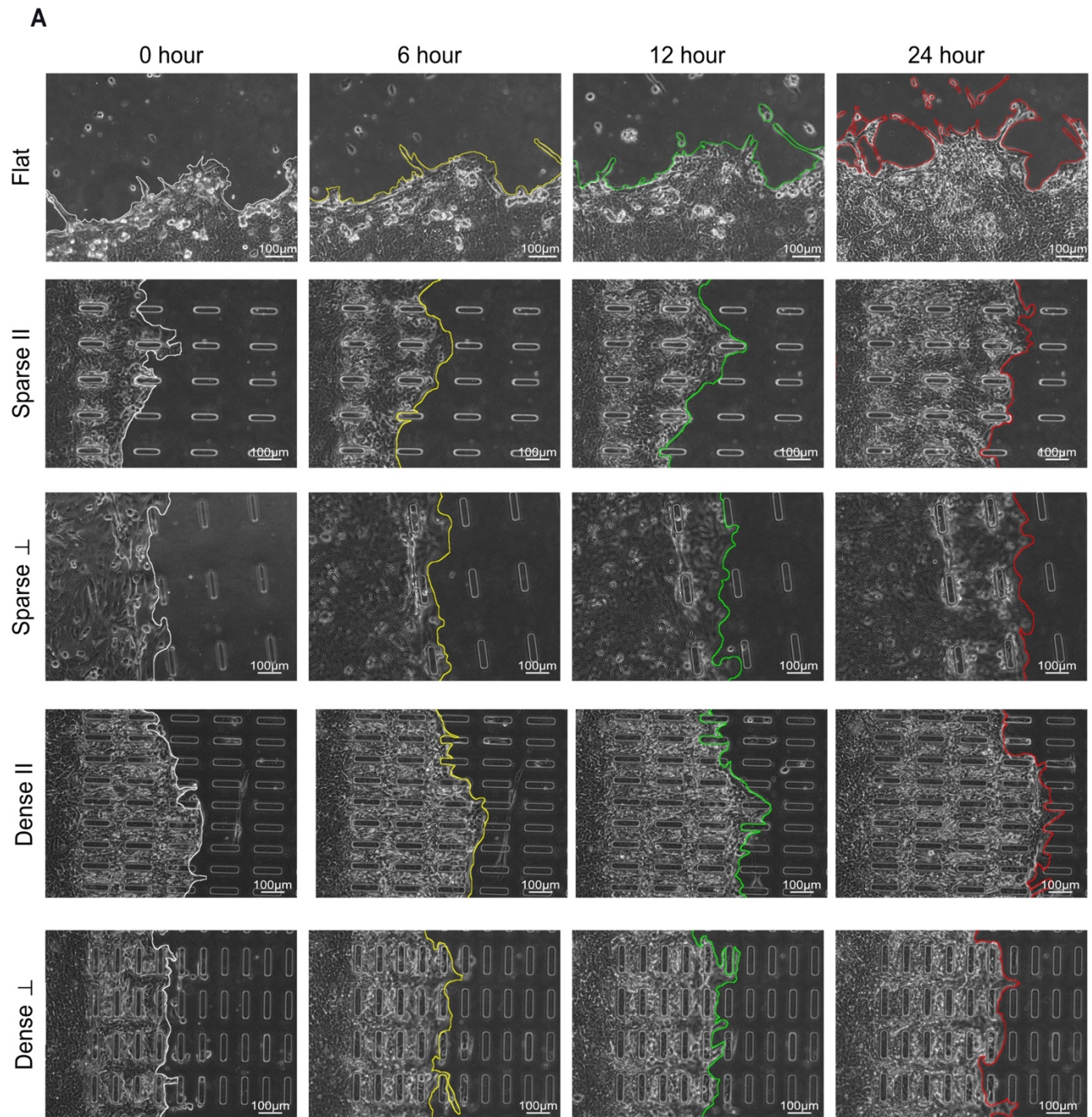


Supplementary Figure 5. AFM workflow and data selection. (A) Representative monolayer 3D projection used to measure cell thickness. Image shows cell nuclei labeled with Hoechst 33342 stain. (B) AFM workflow. Cells throughout the monolayer are indented as described in the main text and Materials and Methods section. As sample thickness can have an effect on cell stiffness as measured by AFM (especially for thin samples), it was accounted for in our data as described previously (top right inset). Representative bad force curves that are ruled out for overall modulus quantitation are shown with reasoning as to why they are ruled out. Upon ruling out data from bad force curves, only data from the remaining good force curves is then compiled and plotted in the heatmaps and Young's Modulus data plots within the main text.

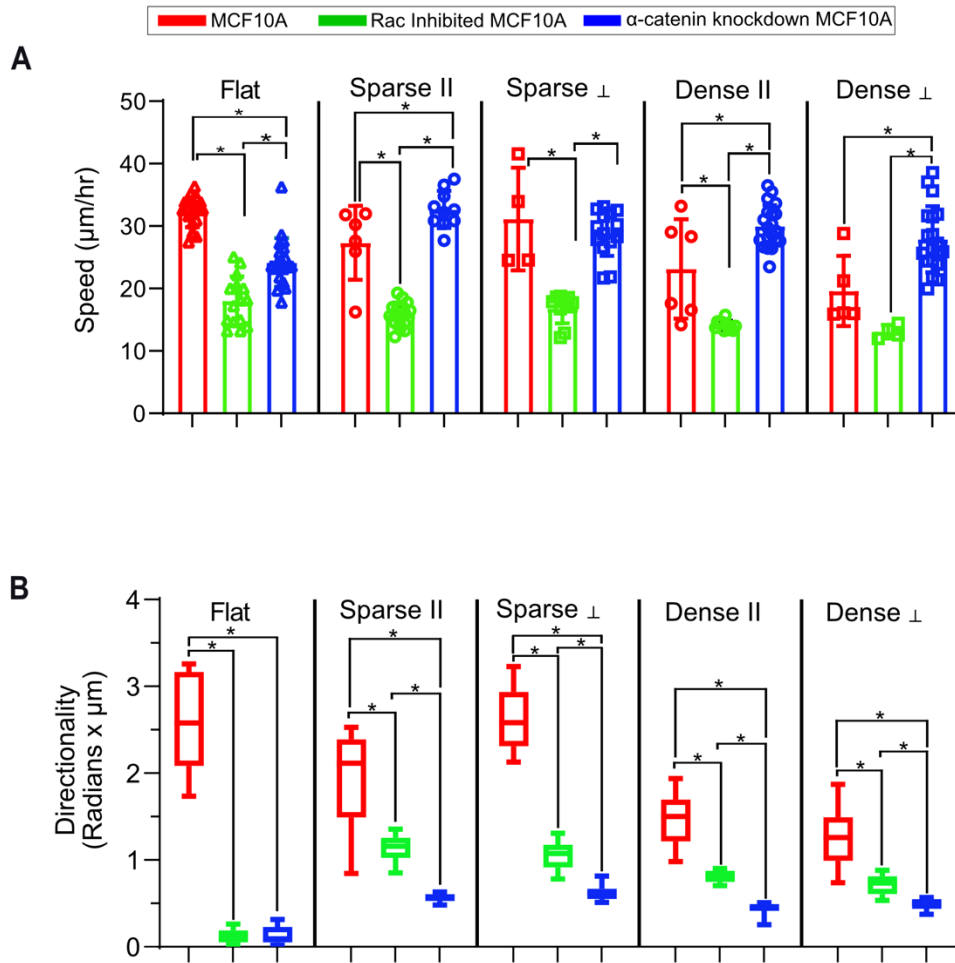


Supplementary Figure 6. Temporal progression of collectively migrating cells through different obstructions and cell-intrinsic properties. Snapshots of simulated cell monolayer at different timepoint ($t = 0, 18, 36, 54$ and $72 H$) for (A) the control/WT condition, (B) after reduced protrusions, and (C) increased leader-like protrusions and lost cell-cell communication. Color coding represents normalized net energy cost.

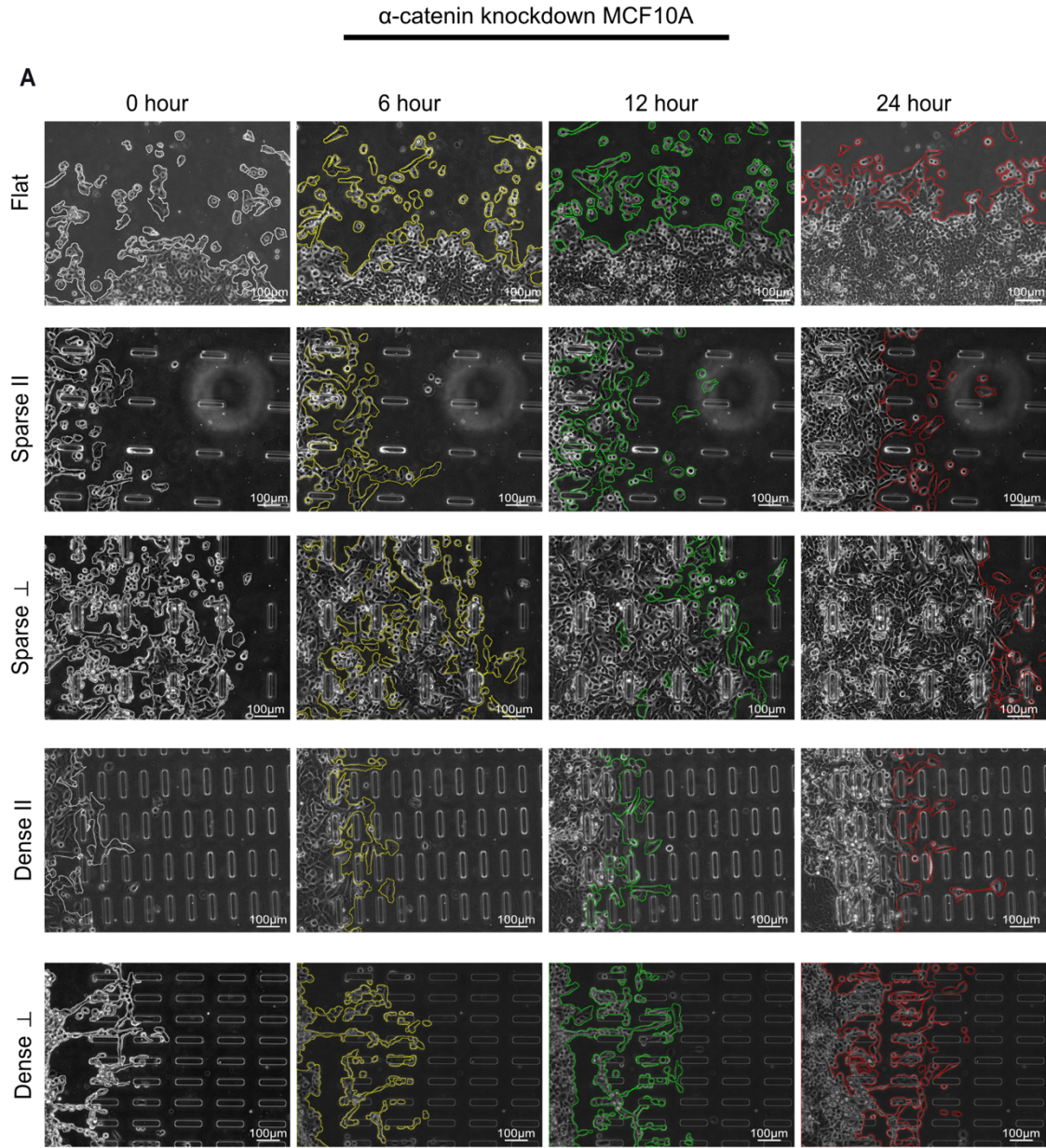
Rac Inhibited MCF10A



Supplementary Figure 7. Leading edge shapes for MCF10A epithelial monolayers after Rac inhibition. Representative phase contrast images of collectively migrating MCF10A cells treated with Rac inhibitor at different timepoints between 0-24hr at 6hr interval, with annotated leading edge shapes, for 5 substrate conditions of varying stump spacing and orientation. Scale bar = 100µm.



Supplementary Figure 8. Speed and directionality comparison after Rac inhibition and α -Catenin knockdown for different obstruction substrates. Migration behavior comparisons for WT MCF10A and after Rac-inhibition and α -Catenin-KD, (A) speed and (B) directionality for all five substrate conditions. Error bars = SD. Significant difference annotated as * $p < 0.05$. $N \geq 4$ fields of view in all plots.



Supplementary Figure 9. Leading edge shapes for MCF10A- α -Catenin-KD epithelial monolayers. Representative phase contrast images of collectively migrating MCF10A cells with depleted α -catenin at different timepoints between 0-24hr at 6hr interval, with annotated leading edge shapes, for 5 substrate conditions of varying stump spacing and orientation. Scale bar = 100 μ m.